

William Hao

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Education

The University of Texas at Austin

Expected May 2027

B.S. Computer Science & B.S. Mathematics – GPA: 4.00/4.00

Relevant Coursework: Data Structures, Discrete Math, Stochastic Processes, Linear Algebra, Probability, Differential Equations

Awards

HackMIT 2025 – 1st Place (Mentra Sponsor Track)

USA Computing Olympiad (USACO) – Gold Division (Java)

Projects

Marvis (HackMIT 2025) • [GitHub](https://github.com)

Sep 2025

- Shipped and demoed an AR “handyman” assistant on MentraOS smart glasses in <24 hours; won the Mentra Sponsor Track.
- Engineered a voice intent parser and state manager for hands-free control of instructions (e.g., “next/previous/repeat”).
- Implemented a backend that reads product barcodes and feeds them into a multi-API AI pipeline (Cerebras, Anthropic, Google Search) to generate and stream step-by-step instructions to the glasses UI.

Tech: TypeScript, Node.js/Bun, MentraOS, Python, Docker

willhao.com • [Live](https://willhao.com) • [GitHub](https://github.com)

Jun 2025 – Present

- Architected and deployed a full-stack portfolio with a strict Next.js/TypeScript environment, featuring live data widgets from Spotify & Chess.com REST APIs with OAuth and server-side caching.
- Achieved a 100/100 PageSpeed score via server-side rendering and an optimized caching strategy, significantly reducing client-side load and API calls.

Tech: TypeScript, Next.js, React, CSS3, Ubuntu, Nginx, PM2

Experience

Texas Undergraduate Computational Finance (UCF) – Analyst

Jan 2025 – Present

Selective student-run organization focused on quantitative finance and algorithmic trading.

- Researched Kelly vs. Markowitz portfolio sizing on S&P sector ETFs (2018 to 2025) using rolling 1-yr windows with transaction cost modeling; extended the analysis to prediction markets via Kalshi and weather APIs. [Pitch deck](#)
- Built a Python backtester (pandas/NumPy) with cost modeling and risk metrics (Sharpe, max drawdown, turnover); implemented VIX mean reversion and moving average strategies on growth stocks (TSLA & NVDA). [Pitch deck](#)
- Authored weekly market reports for 200+ club members and alumni, translating macro trends, earnings analysis, and technical signals into actionable trade ideas. [Sample issue - MSFT](#)

Tech: Python (pandas, NumPy), Jupyter/Colab

UT Math Directed Reading Program (DRP) – Researcher

Jan 2025 – Apr 2025

- Researched dynamical systems and applications (Nilsen, *Randomness and Recurrence in Dynamical Systems*) under PhD mentorship; delivered a math talk on Benford’s Law at end-of-year symposium to 30+ students and faculty. [Presentation](#)

Skills

Programming Languages: Java, Python, C++, TypeScript, JavaScript

Frameworks & Libraries: Next.js, Node.js, React, pandas, NumPy

Developer Tools: Git, Linux, Nginx, PM2

Interests: Competitive chess (USCF 1815; Chess.com 2200), Powerlifting